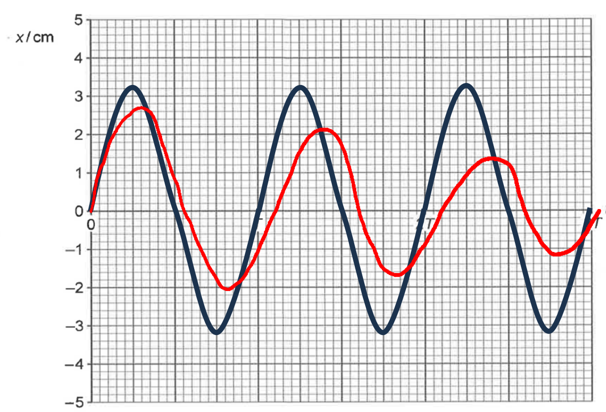
	$Q = 4.57 \times 10$ C
8(a)(i)	The spring constant can be found from the gradient of the force-extension graph within the limits of proportionality: $k = \frac{4-0}{0.2-0} = 20 \text{ N m}^{-1}$

Qn	Solution
8(a)(ii)	$F = kx$ $3.0 = (20)x$ $x = 0.15 \text{ m}$ $E_P = \frac{1}{2}Fx$ $E_P = \frac{1}{2}(3.0)(0.15)$ $E_P = 0.225 \approx 0.23 \text{ J}$
8(a)(iii)	$E_G = mgh$ $E_G = Wx$ $E_G = (3.0)(0.15)$ $E_G = 0.45 \text{ J}$
8(b)	The gravitational potential energy transferred by the mass is not equal to the stored elastic potential energy since part of the gravitational potential energy transferred by the mass is also transferred to the kinetic store of the mass when it stretches the spring and accelerates downwards.
8(c)	Since the mass is displaced from its equilibrium position, it experiences a net force which causes it to accelerate: $F_{net} = ma$ $T - W = ma$ Taking upwards to be positive, and let *e* represent the extension of the spring at equilibrium position (i.e. *ke=mg*) $k(e + x) - mg = ma$ $ke + kx - mg = ma$ $kx = ma$ $a = \frac{kx}{m}$ Since the acceleration and displacement are in opposite directions, $a = -\frac{k}{m}x$
8(d)(i)	Since object is undergoing SHM, $a = -\frac{k}{m}x = -\omega^2 x$

Qn	Solution
	$x = x_0 \sin(\omega t)$ $x = 3.2\sin(\sqrt{\frac{k}{m}}t)$ $x = 3.2\sin(\sqrt{\frac{20}{(3/9.81)}}t)$ $x = 3.2\sin(8.08t)$ When t= 0.50 s, $x = 3.2\sin(8.08\times 0.50)$ $x = -2.51 \approx -2.5$ cm
8(d)(ii)	$v = \omega\sqrt{(x_0^2 - x^2)}$ $v = 8.08\sqrt{(3.2^2 - 2.5^2)}$ $v = 16.15 \approx 16$ m s^{-1}
8(e)	For an oscillation without any damping, the period and amplitude of oscillation stay constant (see black curve). For lightly damped oscillator, the amplitude of oscillation decreases, and the period of oscillation increases (see red curve). 
8(f)(i)	Since both springs experience the same pulling force of 3 N from the hanging object, the extension of the two-spring combination is twice that of the original one-spring system since both springs extend by the same amount.
8(f)(ii)	For the combined two-spring system, the effective spring constant is half that of the original one-spring system in (c) since for the same amount of tension, the extension of the two-spring system is twice that of a single spring system.

Qn	Solution
	$\frac{k}{m}=\omega^2$ $\frac{k}{m}=\left(\frac{2\pi}{T}\right)^2$ $T=2\pi\sqrt{\frac{m}{k}}$ For *k* that is halved, the period of oscillation would increase by a factor of $\sqrt{2}$.
9(a)(i)	